

Notes: Colliard, Foucault, and Lovo (RFS, 2026)

Algorithmic Pricing and Liquidity in Securities Markets

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Experimental environment rather than field data	3
2.2	The main objects in the market-making game	4
2.3	Why the paper introduces a no-adverse-selection benchmark	4
2.4	Baseline parameterization and experimental design	5
2.5	Key outcomes in the extensions	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Market-making game with adverse selection	5
3.2	The market-making game without adverse selection	6
3.3	Glosten-Milgrom benchmark and the competitive price	6
3.4	Why the Nash benchmark is useful	7
3.5	Q-learning algorithm	7
3.6	Why the algorithm need not converge to the competitive action	8
3.7	Experimental findings in the baseline	8
3.8	The core heuristic: learning depends on the signal-to-noise ratio of undercutting	9
3.9	Three analytical properties of the mechanism	9
3.10	Strategic uncertainty	10
3.11	Comparative statics and extensions	10
3.12	Hyperparameters and robustness against smarter opponents	11
4	Identification and instruments (what makes the estimates credible)	11
4.1	This is not an IV paper – identification comes from experimental design	11
4.2	Why the no-adverse-selection treatment is especially credible	11
4.3	Why collusion is not the explanation	12

4.4	Why the estimates are not driven by one lucky or unlucky path	12
4.5	Why the benchmark remains the right comparison even with discrete prices	12
4.6	Robustness and what is left unresolved	12
4.7	Main limitations	12
4.8	Relation to the literature	13
5	Tables and figures	13
5.1	Figure 1 and Figure 2: environment and benchmark	13
5.2	Figure 3: baseline learning dynamics	14
5.3	Figure 4 and Figure 5: comparative statics in sigma and Delta-v	14
5.4	Figure 6 and Figure 7: the mechanism and strategic uncertainty	15
5.5	Figure 8 and Figure 9: entry and tick size	15
5.6	Figure 10: asymmetric reactions to symmetric shocks	16
5.7	Figure 11 and Figure 12: two-period learning and rare states	16
6	Short summary	18
7	Conclusion	18
7.1	Core conclusion (what the paper establishes)	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** How do self-learning market-making algorithms affect liquidity, spreads, and competition in securities markets once we move away from the fully rational Nash benchmark?
- **Setting.** The paper studies “algorithmic market makers” (AMs) that use Q-learning to quote prices in a Glosten-Milgrom-style dealer market with adverse selection.
- **Core challenge.** A rational dealer would undercut rivals down to the competitive Glosten-Milgrom price. But a learning algorithm does not observe expected profits directly. It only observes noisy realized profits and experiments less and less over time.
- **Main mechanism.** AMs learn quickly that posting a *much* lower price than a rival can be profitable, but they struggle to learn that *slightly* undercutting a rival is still profitable. The reason is a low signal-to-noise ratio: the gain from undercutting is small, while realized profits remain very noisy.
- **Main finding.** AMs learn to protect themselves against adverse selection, but they do *not* learn to compete all the way down to the Glosten-Milgrom price. In the long run they settle on supra-competitive prices and earn positive realized spreads.
- **Main comparative statics.** AMs become *less* competitive when their profit volatility rises – for example when clients’ liquidity shocks are more dispersed, when the asset payoff is more volatile, or when there is no adverse selection.
- **Why this matters.** Those predictions differ sharply from the standard benchmark. In the Glosten-Milgrom equilibrium, realized spreads are zero, quoted spreads fall with more dispersed liquidity trading, and no-adverse-selection spreads are zero.
- **Main empirical/experimental strategy.** The paper compares the Nash benchmark to large computational experiments: $K = 1,000$ simulations, each with $T = 1,000,000$ episodes, under a common environment but stochastic trading histories.
- **Policy and market-design implication.** The effects of entry, tick size, and shocks to trading conditions depend on how learning algorithms experiment. In particular, smaller ticks need not tighten spreads when prices are set by self-learning algorithms.
- **Economic takeaway.** The paper is not saying that algorithms “collude.” It is saying that trial-and-error learning with limited experimentation can generate persistent rents and noncompetitive liquidity provision even in a simple Bertrand-style environment.

2 Data and measurement (key objects)

2.1 Experimental environment rather than field data

- **Nature of the evidence.** The paper is primarily an experimental/theoretical study. The main evidence comes from computational experiments in a stylized securities-market environment, not from reduced-form regressions on market data.
- **Unit of analysis.** One *episode* is one trading interaction between one client and N dealers. In the baseline, there is one client and $N = 2$ dealers.

- **What is observed inside each episode.** The algorithms observe the price they post, whether trade occurs, the realized asset payoff, and their realized profit.
- **What is *not* known to dealers.** Dealers do not know the distribution of asset payoffs, the distribution of client demand, or the structure of the game. They only learn from realized outcomes.
- **Why this setup is useful.** It isolates the learning problem cleanly. Any difference between the Nash benchmark and the AM outcomes comes from the way algorithms learn, not from omitted institutional details or ad hoc behavioral assumptions.

2.2 The main objects in the market-making game

- **Asset payoff.** The risky asset has binary payoff $\tilde{v} \in \{v_L, v_H\}$, with

$$\Delta v \equiv v_H - v_L \geq 0, \quad \mu \equiv \Pr(\tilde{v} = v_H).$$

- **Client valuation with adverse selection.** The client's valuation is

$$\tilde{v}_C = \tilde{v} + \tilde{L},$$

where $\tilde{L} \sim N(0, \sigma^2)$ is an independent liquidity shock.

- **Meaning of σ .** A larger σ means the client's private valuation is more dispersed, so trade is more likely at any fixed price and realized dealer profits become more volatile.
- **Trade indicator.** If the best ask is $a_{\min}$, the client buys iff $a_{\min} \leq \tilde{v}_C$.
- **Quoted spread.** The paper measures

$$QS \equiv \mathbb{E}[a_{\min} - \tilde{v}].$$

In practice this is the average ask minus the unconditional expected asset value.

- **Realized spread.** The paper also studies

$$RS \equiv \mathbb{E}[a_{\min} - \tilde{v} \mid \tilde{v}_C > a_{\min}],$$

i.e. expected profit per share *conditional on trade*.

- **Greedy price.** For each AM, the greedy price is the price with the highest current Q-value. This is the algorithm's currently preferred quote.

2.3 Why the paper introduces a no-adverse-selection benchmark

- **Alternative environment.** To isolate adverse selection cleanly, the paper also studies a benchmark in which

$$\tilde{v}_C = \tilde{w}_C + \tilde{L},$$

where $\tilde{w}_C$ is independent of $\tilde{v}$ but has the same unconditional distribution as $\tilde{v}$.

- **Why this is clever.** This construction keeps the unconditional distribution of client valuations unchanged while removing the informational content of order flow.
- **What it buys.** Differences between the adverse-selection and no-adverse-selection treatments can then be attributed to the information content of trading, not to a mechanical change in demand.

2.4 Baseline parameterization and experimental design

- **Baseline primitives.** The paper sets

$$v_H = 4, \quad v_L = 0, \quad \Delta v = 4, \quad \mu = 0.5, \quad \sigma = 5, \quad N = 2.$$

- **Price grid.** AMs choose among 139 possible ask prices from 1.1 to 14.9 in increments of 0.1.
- **Initial Q-values.** Each Q-entry is initialized i.i.d. uniformly on $[3, 6]$. This puts initial values above the maximum one-period payoff so every action gets tried sufficiently often early on.
- **Learning parameters.** In the baseline,

$$\alpha = 0.01, \quad \beta = 8 \times 10^{-5}, \quad \epsilon_t = e^{-\beta t}.$$

- **Simulation size.** The authors run $K = 1,000$ experiments, each lasting $T = 1,000,000$ episodes.
- **Exploration intensity.** With the baseline β , each AM experiments about 12,500 times in expectation – roughly 90 explorations per price on average.
- **Convergence check.** The distribution of prices is essentially stable after about 500,000 episodes, based on Kolmogorov-Smirnov tests across horizons.

2.5 Key outcomes in the extensions

- **Entry extension.** The key objects are average quoted and realized spreads as the number of competing AMs rises from $N = 2$ to $N = 10$.
- **Tick-size extension.** The key objects are quoted and realized spreads at different tick sizes, both with baseline experimentation and with experimentation rescaled so each price gets tried equally often on average.
- **Shock extension.** The key object is the time path of the quoted spread after temporary positive or negative shocks to adverse-selection costs.
- **Two-period extension.** The paper tracks the average first-period price $\bar{a}_1$, the average second-period price after a trade $\bar{a}_{2T}$, after no trade $\bar{a}_{2NT}$, and the unconditional average second-period price $\bar{a}_2$.

Interpretation. Even though the paper is “experimental,” the objects are very economic: spreads, rents, state dependence, and comparative statics against a clear theoretical benchmark.

3 Models and empirical specifications (all equations + interpretation)

3.1 Market-making game with adverse selection

Each dealer n posts an ask a_n . Let $a_{\min} = \min_n \{a_n\}$ be the best offer. If the client buys, demand is split equally among dealers posting the best ask. Dealer n ’s realized trading volume is

$$I(a_n, \bar{a}, \tilde{v}_C) = V(a_{\min}, \tilde{v}_C) Z(a_n, \bar{a}), \quad (1)$$

where $V(a_{\min}, \tilde{v}_C) = 1$ if trade occurs and 0 otherwise, and

$$Z(a_n, \bar{a}) = \begin{cases} 1/N_{\min}, & \text{if } a_n = a_{\min}, \\ 0, & \text{otherwise.} \end{cases}$$

Dealer n 's realized profit is

$$\Pi(a_n, \bar{a}, \tilde{v}_C, \tilde{v}) = I(a_n, \bar{a}, \tilde{v}_C) (a_{\min} - \tilde{v}). \quad (2)$$

The probability of a buy, conditional on payoff v , is

$$D(a_{\min}, v) \equiv \Pr(a_{\min} \leq \tilde{v}_C \mid \tilde{v} = v) = 1 - G(a_{\min} - v), \quad (3)$$

where G is the cdf of $\tilde{L}$.

Interpretation.

- A client is more likely to buy when the asset is actually worth more.
- Therefore trade is bad news for dealers on average: the model embeds classical adverse selection.
- This gives the paper a tight microstructure benchmark: if AMs behave strangely, we can compare them to Glosten-Milgrom rather than to a vague notion of “competition.”

3.2 The market-making game without adverse selection

In the no-adverse-selection environment, the client's valuation is

$$\tilde{v}_C = \tilde{w}_C + \tilde{L},$$

with $\tilde{w}_C$ independent of $\tilde{v}$ and distributed like $\tilde{v}$.

Interpretation.

- Trade no longer reveals information about the asset payoff.
- Yet the unconditional distribution of client valuations is unchanged.
- This makes the treatment contrast sharp: any change in spreads reflects the presence or absence of adverse selection, not a change in raw trading demand.

3.3 Glosten-Milgrom benchmark and the competitive price

Dealer n 's expected profit can be written as

$$\bar{\Pi}(a_n, \bar{a}; \mu) = Z(a_n, \bar{a}) \left[\mu D(a_{\min}, v_H) (a_{\min} - v_H) + (1 - \mu) D(a_{\min}, v_L) (a_{\min} - v_L) \right]. \quad (4)$$

The paper rewrites this as

$$\bar{\Pi}(a_n, \bar{a}; \mu) = Z(a_n, \bar{a}) \mathbb{E}_\mu[V(a_{\min}, \tilde{v}_C)] \left[(a_{\min} - \mathbb{E}_\mu \tilde{v}) - \Delta v \frac{(1 - \mu) \mu \Delta D(a_{\min})}{\mathbb{E}_\mu[V(a_{\min}, \tilde{v}_C)]} \right], \quad (5)$$

where $\Delta D(a_{\min}) = D(a_{\min}, v_H) - D(a_{\min}, v_L)$.

The Glosten-Milgrom price a^* is the lowest price such that dealers make zero expected profit:

$$a^* = \mathbb{E}_\mu(\tilde{v} \mid a^* \leq \tilde{v}_C) = \mathbb{E}_\mu(\tilde{v}) + \Delta v \frac{(1 - \mu)\mu\Delta D(a^*)}{\mathbb{E}_\mu[V(a^*, \tilde{v}_C)]}. \quad (6)$$

Interpretation.

- Equation (5) decomposes expected profit into expected trading volume times expected profit per traded share.
- The bracketed term is quoted spread minus adverse-selection cost.
- In equilibrium, competition pushes that term to zero, so realized spreads are zero.
- In the no-adverse-selection benchmark, $a^* = \mathbb{E}_\mu \tilde{v}$, so both quoted and realized spreads are zero.

3.4 Why the Nash benchmark is useful

A fully rational dealer facing this finitely repeated environment would choose a policy that solves

$$\max \mathbb{E} \left[\sum_{t=1}^T \pi_{n,t} \right]. \quad (7)$$

Because the one-shot market-making game has a unique Nash equilibrium, rational dealers would simply quote the Glosten-Milgrom price each period.

Interpretation.

- The paper is not asking whether the Nash equilibrium exists. It does.
- The question is whether a plausible learning algorithm *finds* it.
- The entire paper is therefore a comparison between rational benchmark behavior and algorithmic learning behavior under the same primitives.

3.5 Q-learning algorithm

Each AM has a Q-vector $Q_{n,t} \in \mathbb{R}^{M \times 1}$. Entry $q_{m,n,t}$ is the algorithm's estimate of the value of quoting price a_m .

The greedy price is the index with the highest Q-value:

$$m_{n,t}^* = \arg \max_m q_{m,n,t-1}, \quad a_{n,t}^* = a_{m_{n,t}^*}.$$

The algorithm explores with probability

$$\epsilon_t = e^{-\beta t}$$

and otherwise exploits by playing the greedy price.

After realized profit $\pi_{n,t}$, the chosen action is updated by

$$q_{m,n,t} = \begin{cases} \alpha\pi_{n,t} + (1 - \alpha)q_{m,n,t-1}, & \text{if } a_{n,t} = a_m, \\ q_{m,n,t-1}, & \text{if } a_{n,t} \neq a_m. \end{cases} \quad (8)$$

Interpretation.

- α is the learning coefficient: how much weight new information gets.
- β controls the speed at which experimentation decays.
- This is the simplest possible RL setup that still captures the exploration/exploitation trade-off.
- The paper deliberately uses a transparent algorithm so the mechanism is economically interpretable.

3.6 Why the algorithm need not converge to the competitive action

The paper emphasizes that the standard convergence guarantees for Q-learning do *not* apply here, for three reasons:

- experimentation is finite in expectation, because ϵ_t decays exponentially;
- the update rule keeps a constant learning rate α , so the weight on new observations does not vanish;
- the environment is not stationary, because each AM's payoff depends on the evolving behavior of the other AM.

Interpretation. This matters because the paper is not a numerical mistake. It is a case where the economics of the environment and the design of the algorithm jointly imply that full convergence to Nash should not be expected.

3.7 Experimental findings in the baseline

In the baseline case:

- the competitive price is $a^* = 2.68$;
- the least competitive pure-strategy Nash price on the discrete grid is 2.8;
- the modal greedy price in the final episode is 4.60;
- the mean greedy price is 4.97;
- the across-experiment standard deviation of the final greedy price is 0.73.

At price 5, each AM's true expected profit is 0.30. By undercutting to 4.9, it could raise expected profit to 0.59, yet the algorithm typically does not learn to do so.

Interpretation.

- AMs clearly learn *something*: prices fall sharply early in training.
- But they stop too early.

- The key empirical fact is not that AMs fail completely, but that they stop at positive realized spreads and leave money on the table.

3.8 The core heuristic: learning depends on the signal-to-noise ratio of undercutting

The paper’s central mechanism is easiest to see by comparing the gain from undercutting to the noise in realized profits. If AM2 posts a_2 , the expected gain for AM1 from undercutting by one tick is

$$\mathbb{E}[\Pi(a_2 - \text{tick}, a_2) - \Pi(a_2, a_2)],$$

and the variance of that gain is approximately

$$\text{Var}[\Pi(a_2 - \text{tick}, a_2)] + \text{Var}[\Pi(a_2, a_2)].$$

In the baseline example where both AMs sit at price 5:

- matching price 5 yields expected profit 0.30;
- undercutting to 4.9 yields expected profit 0.59;
- but realized-profit variance is 1.78 when undercutting and 0.45 when matching.

Interpretation.

- The expected gain from undercutting is real.
- But the realized feedback is so noisy that a few unlucky draws can convince the algorithm that undercutting is bad.
- As prices get closer to competitive levels, the gain from undercutting becomes smaller, so much more experimentation is needed to learn that undercutting is profitable.
- Unfortunately experimentation is decaying exactly when learning becomes hardest.

3.9 Three analytical properties of the mechanism

The paper computes realized-profit variances analytically in the appendix and highlights three key properties:

- If an AM posts a price *above* its rival, it never trades, so profit variance is zero. This is why algorithms quickly learn not to be too expensive.
- For $a_1 \leq a_2$, the profit variance is lower with adverse selection than without adverse selection:

$$\text{Var}_{as}(a_1, a_2) < \text{Var}_{n.as}(a_1, a_2).$$

- Profit variance rises with both σ and Δv .

Interpretation.

- This explains why AMs earn *higher* rents when there is *no* adverse selection: with no adverse selection, the feedback about undercutting is noisier.
- It also explains why realized spreads rise with more dispersed client valuations and with more volatile asset payoffs.

3.10 Strategic uncertainty

Even holding primitives fixed, AMs' payoffs are nonstationary because rival quotes change stochastically over time. The paper calls this *strategic uncertainty*.

To isolate it, the authors let AM2 post a constant price of 5 every period. Then AM1 learns to undercut much faster:

- the average greedy price reaches 4.9 after about 46,369 episodes;
- the modal long-run greedy price becomes 4.9 rather than around 4.6–5.0.

Interpretation. Profit volatility comes from both the environment *and* the rival's learning. Both sources slow competition.

3.11 Comparative statics and extensions

(i) Entry. When the number of AMs rises, undercutting becomes more attractive because tying at a given price splits demand across more dealers, so capturing the whole market is more valuable. At the same time the variance of profits when tied shrinks. The paper predicts and shows that quoted and realized spreads fall with entry.

(ii) Tick size. A smaller tick has two opposing effects:

- it makes undercutting more profitable at a given tied price;
- but it increases the number of actions on the grid, so each action gets explored less often.

With the baseline experimentation rate fixed, quoted and realized spreads are hump-shaped in tick size and can *increase* as the tick is reduced. When the experimentation rate is adjusted so each price gets tried equally often on average, smaller ticks once again make markets more competitive.

(iii) Symmetric shocks with asymmetric responses. The paper shocks adverse-selection costs temporarily after training by changing Δv from 4 to either 7 or 1 for 1,000 episodes. A negative-profit shock (higher adverse selection) causes quoted spreads to jump up quickly, from roughly 3 to just below 4. A positive-profit shock (lower adverse selection) leaves spreads essentially unchanged.

(iv) Two-period order flow. In the two-client extension, the second-period quote depends on the first-period outcome. AMs raise prices after a buy and lower them after no trade, which is directionally correct. But they overreact after buys and underreact after no trade. As a result the average second-period price is *higher* than the first-period price, opposite to the Glosten-Milgrom benchmark.

Interpretation.

- The paper keeps finding the same principle: rare states and noisy states are the least competitive ones.
- This is why the algorithms become especially noncompetitive in the second period after a trade, because those states are visited less often and the Q-matrix becomes harder to learn.

3.12 Hyperparameters and robustness against smarter opponents

The paper asks whether dealer competition over algorithm design fixes the problem. Dealers are allowed to choose

$$\alpha \in \{0.001, 0.01, 0.1\}, \quad \beta \in \{5 \times 10^{-6}, 8 \times 10^{-5}, 3.2 \times 10^{-4}\}.$$

The main findings are:

- only a handful of hyperparameter pairs are “stable” in the sense that unilateral deviations do not look significantly more profitable;
- the baseline pair (α_m, β_m) is one of these stable configurations;
- there is no race to the top in experimentation, because more experimentation helps undercut today but lowers profits tomorrow once the rival adapts.

The paper also studies competition from an “omniscient” Bayesian dealer in the Internet Appendix and argues that such a dealer does not want to undercut mechanically forever. Instead, the sophisticated player often prefers to preserve high aggregate profits and capture most of them.

Interpretation. The AMs’ supra-competitive prices are robust. They are not obviously competed away by simple parameter choices or by a single sophisticated opponent.

4 Identification and instruments (what makes the estimates credible)

4.1 This is not an IV paper – identification comes from experimental design

The paper does not identify a causal effect with instruments in the econometric sense. Instead, it identifies the role of learning by comparing:

- a fully rational benchmark (the Glosten-Milgrom/Nash equilibrium),
- and algorithmic outcomes in the exact same environment.

Interpretation. The identification logic is experimental: hold the economic environment fixed, change the behavior-generating process, and study the resulting comparative statics.

4.2 Why the no-adverse-selection treatment is especially credible

A naive way to “reduce adverse selection” would be to raise σ . But that would also change the whole demand distribution. The paper instead constructs a benchmark where client valuations have the same unconditional distribution but are independent of the asset payoff.

Why this matters.

- It isolates the information content of order flow.
- It makes the comparison with and without adverse selection much cleaner than a simple parameter change in σ would.

4.3 Why collusion is not the explanation

A natural alternative story would be that Q-learning algorithms converge to collusive play, as in some product-market models. The paper argues this is ruled out by design:

- algorithms cannot condition on past prices or past trade outcomes when choosing current actions in the main one-period setup;
- therefore they cannot implement history-dependent punishment strategies.

Interpretation. The AMs are not tacitly colluding in the repeated-game sense. They simply fail to learn to compete.

4.4 Why the estimates are not driven by one lucky or unlucky path

The paper deals with path dependence very directly:

- it runs $K = 1,000$ experiments for each parameterization;
- it reports the distribution of outcomes across experiments, not just one history;
- it focuses on long-run distributions once the outcome distribution has stabilized.

Interpretation. The important object is the *average long-run behavior* of a learning algorithm, not a single simulation trace.

4.5 Why the benchmark remains the right comparison even with discrete prices

Because the price grid is discrete, the exact Glosten-Milgrom price need not lie on the grid, and discrete pricing can generate multiple pure-strategy Nash equilibria for coarse enough ticks. The paper therefore compares simulations to the *least competitive* Nash price on the grid whenever necessary.

Interpretation. This makes the benchmark conservative. If AMs still look uncompetitive relative to the least competitive Nash price, the result is hard to dismiss as a discretization artifact.

4.6 Robustness and what is left unresolved

- The basic results survive alternative values of α and β .
- They also survive algorithmic variations in the Internet Appendix that relax some convergence issues.
- The main unresolved issue is normative and institutional: how dealers choose hyperparameters in practice, and whether they do so online or offline.

4.7 Main limitations

- The main algorithm is deliberately simple Q-learning, not a full modern deep-RL architecture.
- The baseline environment is one-sided: only buyers arrive.

- There are no inventory penalties in the benchmark experiments.
- Only dealers learn; clients do not optimize strategically through learning.
- The main experiments exclude humans, although real markets mix humans and algorithms.

Interpretation. These are genuine limitations, but they mostly make the paper *cleaner*. The core message is about a basic learning friction that is likely to survive in richer settings.

4.8 Relation to the literature

- Relative to the algorithmic-pricing literature, the paper studies learning to compete in a securities-market setting with uncertainty about both demand and costs.
- Relative to market-microstructure theory, the paper keeps the Glosten-Milgrom environment but replaces rational dealers with learning algorithms.
- Relative to learning-in-games theory, the paper is close in spirit to self-confirming equilibrium: players may behave optimally given beliefs that are only correct on the path actually played.
- Relative to market design, the paper shows that standard prescriptions such as reducing tick size need not have standard effects when prices are set by self-learning algorithms.

5 Tables and figures

This paper is almost entirely figure-based in the main text: there are no main tables. The cleanest way to read the evidence is therefore by figure blocks.

5.1 Figure 1 and Figure 2: environment and benchmark

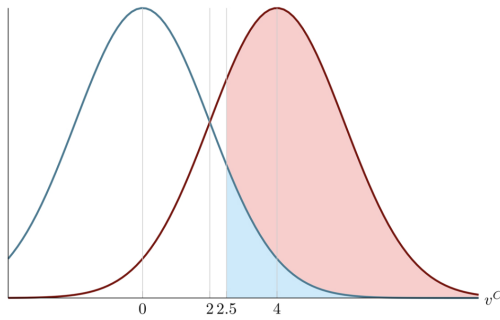


Figure 1
Distribution of the client's valuation, v^C
Parameter values are $v_H = 4$, $v_L = 0$, $\mu = \frac{1}{2}$, and $\sigma = 5$. This figure displays the distributions of clients' valuations for $v = v_H$ (red) and $v = v_L$ (blue). If dealers' best offer is $a^{min} = 2.5$, the likelihood that the client buys the asset is given by (2) the blue area when $v = v_L$ and (2) the sum of the blue and red areas when $v = v_H$.

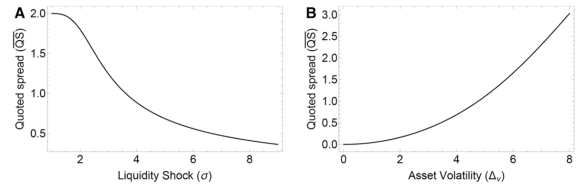


Figure 2
Glosten-Milgrom benchmark
Parameters: $E_\mu(v) = 2$, $\mu = \frac{1}{2}$, $\Delta_v = 4$, $\sigma = 5$. The figure shows the equilibrium quoted spread in the Glosten-Milgrom benchmark as a function of the volatility of the asset payoff Δ_v (panel A) and the variance of clients' liquidity shocks σ (panel B).

Read:

- Figure 1 shows why adverse selection exists: for any fixed price, the client is more likely to buy when the asset payoff is high than when it is low.

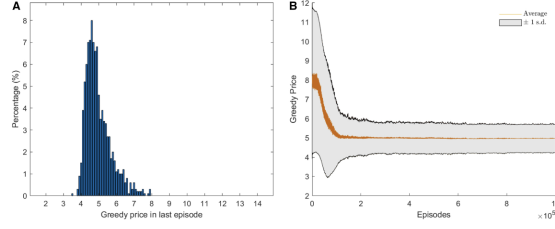


Figure 3
Greedy price of AM, in the adverse selection case
 Parameters: $\sigma = 5$, $\Delta v = 4$, $N = 2$, $\mu = \frac{1}{2}$, $E(t) = 2$, $T = 1,000,000$, and $K = 1,000$. Panel A shows a histogram of the greedy price of AM_{*i*} in episode T . For each possible price a between 1.10 and 14.90, the bar indicates the percentage of the 1,000 experiments in which $a_{i,T}^* = a$. Panel B shows for each episode t between 1 and 1,000,000 the average of AM_{*i*}'s greedy price $a_{i,t}^*$ across the 1,000 experiments conducted. As a measure of dispersion, we also compute the standard deviation of $a_{i,t}^*$ across experiments and plot the average of $a_{i,t}^*$ plus/minus one standard deviation (with a 500-episode moving average for better readability).

- Figure 2 summarizes the Glosten-Milgrom benchmark. With adverse selection, the quoted spread rises with Δv and falls with σ ; without adverse selection, both quoted and realized spreads are zero.
- These figures are important because they pin down what the AMs *should* do if they learned the Nash equilibrium.

5.2 Figure 3: baseline learning dynamics

Read:

- Panel A shows the distribution of the final greedy price across experiments. The mass is well above the competitive benchmark.
- Panel B shows the training dynamics: AMs cut prices quickly at first, then the decline slows and essentially stops.
- The figure is the paper's core empirical fact: AMs learn a lot, but not enough.

5.3 Figure 4 and Figure 5: comparative statics in sigma and Delta-v

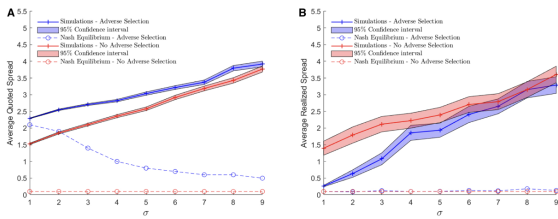


Figure 4
Average quoted spread $\bar{Q}$ and average realized spread $\bar{R}$ in the adverse selection case and the no adverse selection case for different values of the dispersion of clients' liquidity shocks σ
 Other parameters: $\Delta v = 4$, $N = 2$, $\mu = \frac{1}{2}$, $E(t) = 2$, $T = 1,000,000$, and $K = 1,000$. Panel A plots the average over 1,000 experiments of the quoted spread, with 95% confidence intervals, both in the adverse selection case and in the no adverse selection case. The graph additionally plots the values of the quoted spread in both cases in the Glosten-Milgrom benchmark of Section 2.3 (accounting for price discretization). Panel B plots the average over 1,000 experiments of the realized spread, with 95% confidence intervals, both in the adverse selection case and in the no adverse selection case. The graph additionally plots the values of the realized spread in both cases in the Glosten-Milgrom benchmark of Section 2.3 (accounting for price discretization).

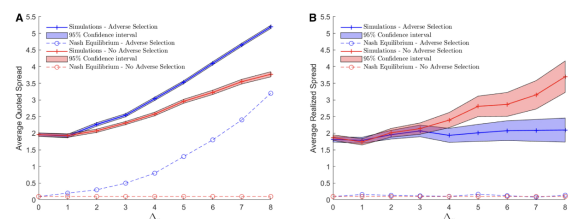


Figure 5
Average quoted spread $\bar{Q}$ and average realized spread $\bar{R}$ in the adverse selection case and the no adverse selection case for different values of the asset volatility Δv
 Other parameters: $\sigma = 5$, $N = 2$, $\mu = \frac{1}{2}$, $E(t) = 2$, $T = 1,000,000$, and $K = 1,000$. Panel A plots the average over 1,000 experiments of the quoted spread, with 95% confidence intervals, both in the adverse selection case and in the no adverse selection case. The graph additionally plots the values of the quoted spread in both cases in the Glosten-Milgrom benchmark of Section 2.3 (accounting for price discretization). Panel B plots the average over 1,000 experiments of the realized spread, with 95% confidence intervals, both in the adverse selection case and in the no adverse selection case. The graph additionally plots the values of the realized spread in both cases in the Glosten-Milgrom benchmark of Section 2.3 (accounting for price discretization).

Read:

- Figure 4 shows that both quoted and realized spreads rise with the dispersion of clients' liquidity shocks σ .
- Figure 5 shows that both quoted and realized spreads also rise with the volatility of the asset payoff Δv .
- In both figures, AMs charge positive realized spreads even when there is *no* adverse selection.

- In both figures, realized spreads are smaller with adverse selection than without.

Why it matters.

- These are exactly the comparative statics the noisy-learning mechanism predicts.
- They are also exactly the patterns the standard benchmark does *not* predict.

5.4 Figure 6 and Figure 7: the mechanism and strategic uncertainty

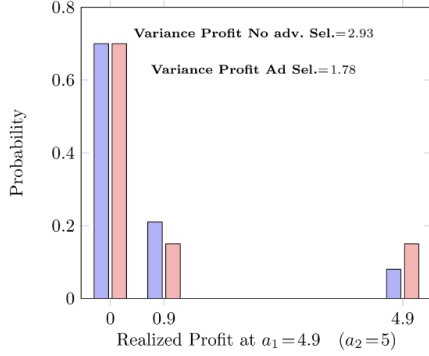


Figure 6
Distribution of profits with and without adverse selection
The figure compares the distribution of realized profits for AM_1 when it posts a price of 4.9 while AM_2 posts a price of 5.0 in the baseline case ($\mu = 0.5$, $v_H = 4$, $v_L = 0$; $\sigma = 5$), in the case without adverse selection (red), and in the case with adverse selection (blue).

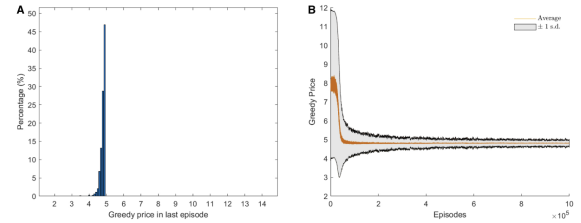


Figure 7
Greedy price of AM_1 when AM_2 plays a constant price in the adverse selection case
Parameters: $\sigma = 5$, $a_0 = 4$, $N = 2$, $\mu = \frac{1}{2}$, $E(\epsilon) = 2$, $T = 1,000,000$, and $K = 1,000$. AM_2 plays a constant price of 5.0 in every episode, while AM_1 uses a Q-learning algorithm with $\alpha = 0.01$ and $\beta = 0.00008$. Panel A shows a histogram of the greedy price of AM_1 in episode T ; for each possible price a between 1.10 and 14.90 the bar indicates the percentage of the 1,000 experiments in which $a_1^g = a$. Panel B shows for each episode t the average of AM_1 's greedy price a_1^g across the 1,000 experiments conducted. As a measure of dispersion, we also compute the standard deviation of a_1^g across experiments and plot the average of a_1^g plus/minus one standard deviation (with a 500-episode moving average for better readability).

Read:

- Figure 6 compares the distribution of realized profits from undercutting in the adverse-selection and no-adverse-selection cases. It visualizes the paper's key point that no-adverse-selection profits are more volatile.
- Figure 7 freezes AM_2 at a constant price of 5 and shows that AM_1 undercuts faster and more aggressively than in the full two- AM experiment.

Why it matters.

- Figure 6 makes the signal-to-noise mechanism concrete.
- Figure 7 shows that part of the learning failure is due to strategic uncertainty generated by the rival's own experimentation.

5.5 Figure 8 and Figure 9: entry and tick size

Read:

- Figure 8 shows that more AM s lead to lower quoted and realized spreads.
- Figure 9 shows the nonstandard tick-size result. With baseline experimentation fixed, spreads are hump-shaped in the tick size and can widen when the tick is reduced. When experimentation is rescaled so each price is explored equally often on average, smaller ticks lower spreads in the standard way.

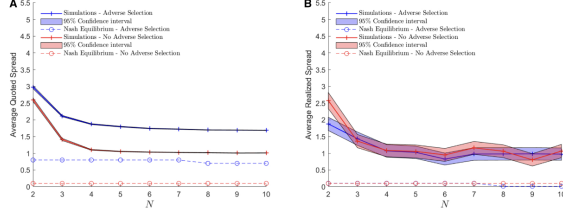


Figure 8
Average quoted spread $\bar{Q}$ and average realized spread $\bar{R}$ in the adverse selection case and the no adverse selection case for different values of the number N of AMs. Other parameters: $\sigma = 5$, $\Delta_0 = 4$, $\mu = \frac{1}{2}$, $E(v) = 2$, $T = 1,000,000$, and $K = 1,000$. Panel A plots the average over 1,000 experiments of the quoted spread, with 95% confidence intervals, both in the adverse selection case and in the no adverse selection case. The graph additionally plots the values of the quoted spread in both cases in the Glosten-Milgrom benchmark of Section 2.3 (accounting for price discreteness). Panel B plots the average over 1,000 experiments of the realized spread, with 95% confidence intervals, both in the adverse selection case and in the no adverse selection case. The graph additionally plots the values of the realized spread in both cases in the Glosten-Milgrom benchmark of Section 2.3 (accounting for price discreteness).

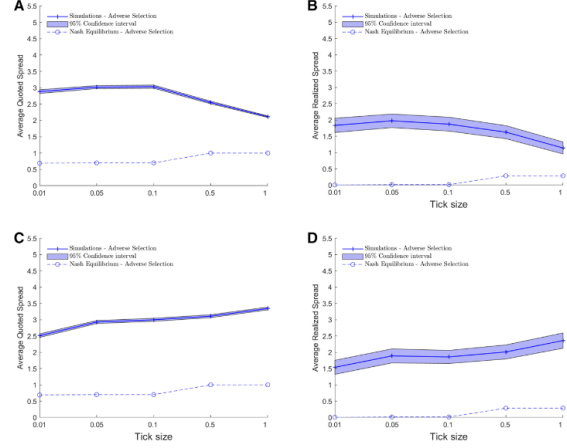


Figure 9
Average quoted spread $\bar{Q}$ and average realized spread $\bar{R}$ in the adverse selection case for different values of the tick size. Other parameters: $\sigma = 5$, $\Delta_0 = 4$, $N = 2$, $\mu = \frac{1}{2}$ and $E(v) = 2$. We report the average over $K = 1,000$ experiments of the quoted spread (panel A and panel C) and the realized spread (panel B and panel D) in episode T . In panels A and B we use the baseline value $\beta = 8 \cdot 10^{-5}$. In panels C and D we adjust β to the tick size so the average number of experimentations per price in the grid is constant (see the main text). The graphs additionally plot the values of the quoted spreads and the realized spreads in the Glosten-Milgrom benchmark of Section 2.3 (accounting for price discreteness).

Why it matters.

- Figure 8 says entry improves learning because undercutting becomes more informative.
- Figure 9 is the market-design punchline: finer pricing need not make markets more competitive when algorithms have limited exploration budgets.

5.6 Figure 10: asymmetric reactions to symmetric shocks

Read:

- The figure applies temporary positive and negative shocks to adverse-selection costs after the algorithms have already trained.
- AMs respond quickly to a negative-profit shock by widening spreads.
- They barely react at all to a positive-profit shock that should, symmetrically, justify lower spreads.

Why it matters.

- The algorithm already likes its long-run greedy price.
- Positive shocks reinforce that price, while negative shocks can push its Q-value below that of a higher price.
- So learning creates state dependence and asymmetry even when the primitive shock is symmetric.

5.7 Figure 11 and Figure 12: two-period learning and rare states

Read:

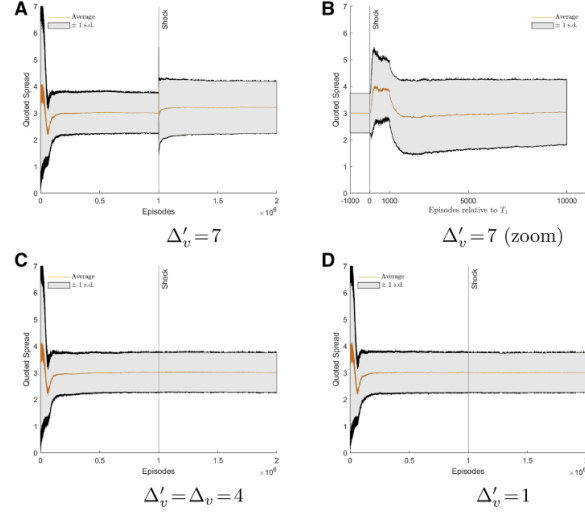


Figure 10
Dynamics of quoted spreads after a shock on adverse selection
Parameters: $\sigma = 5$, $\Delta_v = 4$, $N = 2$, $\mu = \frac{1}{2}$, $E(v) = 2$. Each panel shows the dynamics of the quoted spread, averaged over $K = 1,000$ experiments. Between episodes $T_1 + 1$ and $T_1 + 1,000$, with $T_1 = 10^5$, there is a shock to Δ_v , the value of which is changed to Δ'_v . Δ'_v is equal to 7 in panels A and B (high adverse selection shock), is equal to 1 in panel D (low adverse selection shock), and remains equal to 4 in panel C (Placebo). Panel B zooms in on episodes between $T_1 - 1,000$ and $T_1 + 10,000$, while all other panels show all episodes between 1 and $T = 2 \cdot 10^6$.

- Figure 11 shows that AMs raise second-period prices after a first-period trade and lower them after no trade, but they overreact in the first case and underreact in the second.
- Figure 12 shows the striking unconditional implication: the average second-period price exceeds the average first-period price, opposite to the Glosten-Milgrom benchmark.

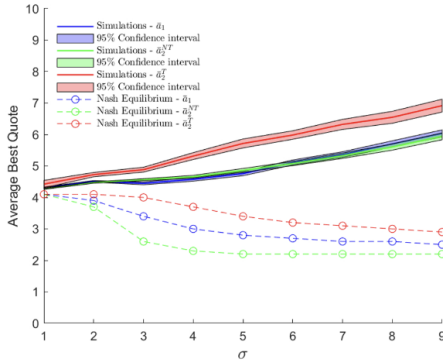


Figure 11
Average first-period price $\bar{a}_1$ and second-period price after a trade $\bar{a}_2^T$ and after no trade $\bar{a}_2^{NT}$ for different values of the dispersion of clients' liquidity shocks σ
Other parameters: $\Delta_v = 4$, $N = 2$, $\mu = \frac{1}{2}$, $E(v) = 2$, $T = 1,000,000$, and $K = 1,000$. This graph plots the average over 1,000 experiments of the first-period and second-period prices, with 95% confidence intervals. The graph additionally plots the values of these prices in the Glosten-Milgrom benchmark (accounting for price discreteness).

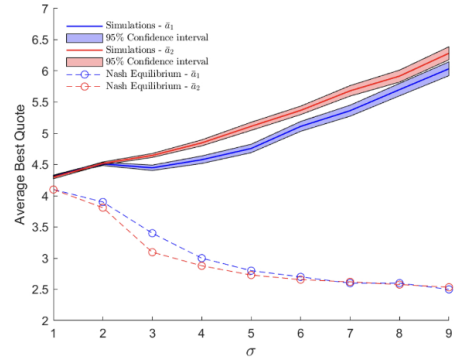


Figure 12
Average first-period price $\bar{a}_1$ and average second-period price $\bar{a}_2$ for different values of the dispersion of clients' liquidity shocks σ
Other parameters: $\Delta_v = 4$, $N = 2$, $\mu = \frac{1}{2}$, $E(v) = 2$, $T = 1,000,000$, and $K = 1,000$. This graph plots the average over 1,000 experiments of the first-period and second-period price, with 95% confidence intervals. The graph additionally plots the values of these prices in the Glosten-Milgrom benchmark (accounting for price discreteness). Average prices in both periods decrease with σ in the Glosten-Milgrom benchmark and increase with σ in the simulations. Prices are on average lower in the second period than in the first period in the Glosten-Milgrom benchmark, whereas the opposite obtains in the simulations.

Why it matters.

- The more states the algorithm must learn about, the less data it has in each state.
- Rarely visited states are the least competitive ones.
- This is a direct, testable field prediction.

6 Short summary

- **Method.** The paper embeds Q-learning dealers in a Glosten-Milgrom dealer market and compares their long-run behavior to the Nash benchmark using large-scale computational experiments.
- **Main conceptual result.** Learning to avoid adverse selection is much easier than learning to compete. The hard part is discovering that a *slight* undercut remains profitable when realized feedback is noisy and experimentation is fading.
- **Main quantitative result.** In the baseline, the long-run greedy price averages about 4.97 even though the competitive price is only 2.68.
- **Main comparative-statics result.** Higher profit volatility leads to wider spreads and higher rents, even in environments where the standard benchmark predicts the opposite.
- **Main policy lesson.** Entry generally helps, but reducing tick size can backfire unless algorithms are reparametrized to experiment enough on the finer price grid.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that Q-learning market makers can behave realistically in some dimensions – they charge more when adverse selection is stronger and update in the right direction after order flow – while still failing to learn competitive pricing.
- It also establishes that this failure is systematic, not accidental: it is generated by noisy feedback and decaying experimentation.
- Finally, it shows that this learning friction leaves empirical footprints that differ sharply from the standard Nash equilibrium.

7.2 Economic intuition

- Undercutting is easiest to learn when prices are far above competitive levels because the gain is large relative to the noise.
- As prices get closer to competitive levels, the gain from undercutting shrinks but realized-profit variance remains large.
- The algorithm then runs out of experimentation exactly when it most needs more data, so prices freeze above competitive levels.

7.3 Takeaway

This paper shows that market quality in AI-driven markets cannot be understood by looking only at the static game being played. One must also study the learning technology. In this setting, the relevant friction is not market power in the standard sense, nor explicit collusion, but the interaction of noisy realized feedback, limited experimentation, and strategic uncertainty. That mechanism can

generate persistently wide spreads, nonstandard responses to tick-size changes, and especially high rents in rare or noisy states – all of which are testable predictions for modern electronic markets.